

Qualification Pack



Slurry Pump Operator (Mines)

Underground Metal/ Opencast/ Underground Coal/ Rare Earth

QP Code: MIN/Q3207

Version: 1.0

NSQF Level: 3.0

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MIN/Q3207: Slurry Pump Operator (Mines)

Brief Job Description

The individual in this role is responsible for transferring material or slurry from one place to another in the mines or from one location to another location for processing, by operating and maintaining the slurry pump unit.

Personal Attributes

This job requires the individual to concentrate on the job at hand and complete it efficiently and effectively. The individual should be physically agile, strong, have good eye-sight and not suffer from color-blindness.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N3221: Prepare slurry pump unit for operation](#)
2. [MIN/N3222: Perform operation of slurry pump unit](#)
3. [MIN/N3223: Perform maintenance of the slurry pump](#)
4. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Electives(mandatory to select at least one):

Elective 1: Underground Metal

1. [MIN/N1702: Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines](#)

Elective 2: Opencast

1. [MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines](#)

Elective 3: Underground Coal

1. [MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines](#)

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Elective 4: Rare Earth

1. [MIN/N1705: Follow Health, Safety and Environmental guidelines for Rare Earth \(RE\) Chemical plant](#)

Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
Country	India
NSQF Level	3.0
Credits	19
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3134.0300, NCO-2015/3132.1000
Minimum Educational Qualification & Experience	10th grade pass (or equivalent) OR 9th grade pass (with 1.5 years relevant experience in Mining Industry) OR 8th grade pass (with 3 years relevant experience in Mining Industry) OR 5th grade pass (with 7.5 years relevant experience in Mining Industry) OR Previous relevant Qualification of NSQF Level (2.0: Assistant-Opencast Mines or Assistant-Underground Mines with 3 years relevant experience in Mining Industry)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	26/08/2027
NSQC Approval Date	27/08/2024



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MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP



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Version	1.0
Reference code on NQR	QG-03-MG-02927-2024-V1-SCMS
NQR Version	1.0

Remarks:

Total Duration: 390 Hours (360 hours Domain + 30 Hours Employability Skills)

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MIN/N3221: Prepare slurry pump unit for operation

Description

This OS unit is about inspecting & preparing the slurry pump unit for activities that need to be carried out.

Scope

The scope covers the following :

- Prepare for work
- Inspect and prepare the slurry pump unit for operation

Elements and Performance Criteria

Prepare for work

To be competent, the user/individual on the job must be able to:

- PC1.** identify the work requirements from the instructions received from supervisor and interpreting information from previous day/shift logbook
- PC2.** use personal protective equipment appropriate for work activities
- PC3.** select appropriate type of auxiliary equipment for work activities

Inspect and prepare the slurry pump unit for operation

To be competent, the user/individual on the job must be able to:

- PC4.** check various controls, gauges, slurry pump and allied components and accessories for proper functioning and any defect
- PC5.** check the direction of rotation of motor is corresponding to the regulated rotation direction of pump and ensure that motor should be tested alone and it must not be connected with pump during check
- PC6.** ensure that concentricity of motor shaft and slurry pump shaft is correct and elastic cushion of coupling is correct and complete
- PC7.** check the impeller clearance by manually turn the slurry pump (including motor) for detecting any friction, tightness etc. in it
- PC8.** check the bearing oil level in the bearing box and replenish it as per SOP, if requires
- PC9.** check the condition of inlet and outlet valves and ensure that they are working properly and in normal state
- PC10.** check the pipe system and other parts such as anchor bolt, flange sealing gasket and other bolts for proper working and good condition
- PC11.** check all foundation bolts and outlet pipework bolts for appropriate tightness and condition
- PC12.** ensure all electrical, hydraulic, pneumatic, sealant and lubrication systems (as applicable) are connected and operational
- PC13.** add the seal water (cooling water for mechanical seal), before starting the slurry pump
- PC14.** perform pump priming by opening the air vents on the inlet and outlet piping until the pumped fluid flows out and after that closing the air vents
- PC15.** ensure the inlet pipe and pump casing are completely full of liquid before starting continuous duty operation and pump fluid must be above the pump casing

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PC16. maintain the operation and maintenance logbook to detail all activities conducted before starting the operation

PC17. report defects precisely to the supervisor if beyond scope of his/her role

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** job-specific documents e.g. daily maintenance checklist and importance of the same
- KU3.** risk and impact of not following defined procedures/work instructions
- KU4.** cost of equipment and loss for the company that results from damage of equipment
- KU5.** implications of delays in the process
- KU6.** safety guidelines specified by Directorate General of Mines Safety (DGMS)
- KU7.** different types of mines and detail of the mine one is working in
- KU8.** code of practice in specific areas of the mine
- KU9.** importance of safety in the vicinity of machinery
- KU10.** duties of workmen under the Mines Act-1952
- KU11.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU12.** international safety guidelines and policies like COSHH, OSHA etc.
- KU13.** the outcome of violation of safety procedures
- KU14.** different types of slurry pumps
- KU15.** functioning of slurry pump and its various components and accessories
- KU16.** normal running characteristics of slurry pump unit
- KU17.** how to inspect slurry pump unit and its components before operation
- KU18.** emergency response /disaster management plan prepared by the organization
- KU19.** signages, mining area-specific signs and other safety and emergence signals
- KU20.** basic computer work
- KU21.** how to use services such as NEFT, IMPS, UPI, RTGS for online banking
- KU22.** importance of entrepreneurship and characteristics of an effective entrepreneur
- KU23.** various campaigns run by government nationally such as Swachh Bharat Abhiyan, make in India campaign etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** read and interpret symbols and readings
- GS3.** write work related information in English/regional language
- GS4.** communicate the work requirements to the supervisor and co-workers



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- GS5.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS6.** make decisions pertaining to the concerned area of work
- GS7.** plan and organize the work order and tasks
- GS8.** recognise a workplace problem and take suitable action
- GS9.** follow instructions and work on areas of improvement identified
- GS10.** complete the assigned tasks timely
- GS11.** use reasoning skills to identify and resolve basic problems
- GS12.** analyze and detect any potential problems which could arise during operation
- GS13.** report to the supervisor or deal with a colleague individually, depending on the type of concern

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for work</i>	5	8	-	5
PC1. identify the work requirements from the instructions received from supervisor and interpreting information from previous day/shift logbook	1	2	-	1
PC2. use personal protective equipment appropriate for work activities	2	3	-	2
PC3. select appropriate type of auxiliary equipment for work activities	2	3	-	2
<i>Inspect and prepare the slurry pump unit for operation</i>	25	42	-	15
PC4. check various controls, gauges, slurry pump and allied components and accessories for proper functioning and any defect	2	3	-	2
PC5. check the direction of rotation of motor is corresponding to the regulated rotation direction of pump and ensure that motor should be tested alone and it must not be connected with pump during check	2	3	-	1
PC6. ensure that concentricity of motor shaft and slurry pump shaft is correct and elastic cushion of coupling is correct and complete	1	3	-	1
PC7. check the impeller clearance by manually turn the slurry pump (including motor) for detecting any friction, tightness etc. in it	2	3	-	1
PC8. check the bearing oil level in the bearing box and replenish it as per SOP, if requires	2	3	-	1
PC9. check the condition of inlet and outlet valves and ensure that they are working properly and in normal state	2	3	-	1
PC10. check the pipe system and other parts such as anchor bolt, flange sealing gasket and other bolts for proper working and good condition	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. check all foundation bolts and outlet pipework bolts for appropriate tightness and condition	2	3	-	1
PC12. ensure all electrical, hydraulic, pneumatic, sealant and lubrication systems (as applicable) are connected and operational	1	3	-	1
PC13. add the seal water (cooling water for mechanical seal), before starting the slurry pump	2	3	-	1
PC14. perform pump priming by opening the air vents on the inlet and outlet piping until the pumped fluid flows out and after that closing the air vents	2	3	-	1
PC15. ensure the inlet pipe and pump casing are completely full of liquid before starting continuous duty operation and pump fluid must be above the pump casing	1	3	-	1
PC16. maintain the operation and maintenance logbook to detail all activities conducted before starting the operation	2	3	-	1
PC17. report defects precisely to the supervisor if beyond scope of his/her role	2	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3221
NOS Name	Prepare slurry pump unit for operation
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
NSQF Level	3.0
Credits	1
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	26/08/2027
NSQC Clearance Date	27/08/2024

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MIN/N3222: Perform operation of slurry pump unit

Description

This OS unit is about performing operation of slurry pump unit in the mines for transferring the slurry from one place to another place.

Scope

The scope covers the following :

- Start and operate the slurry pump unit
- Stop the slurry pump unit

Elements and Performance Criteria

Start and operate the slurry pump unit

To be competent, the user/individual on the job must be able to:

- PC1.** ensure flushing and/or cooling liquid supplies are turned on and all vent connections are closed before starting the pump
- PC2.** open the inlet valve and fully close or partially open the outlet valve before starting the pump
- PC3.** open any recirculation or cooling lines
- PC4.** start the motor and slowly open the outlet valve until the pump reaches the desired flow
- PC5.** check the pressure gauge to ensure that the pump reaches the required outlet pressure
- PC6.** stop the pump, if there is no pressure or low pressure
- PC7.** perform the pump priming process again and restart the motor again
- PC8.** monitor the pump for bearing temperature, excessive vibration, and noise during the operation
- PC9.** shut down the pump immediately, if the pump exceeds normal levels and take the corrective action as per SOP
- PC10.** repeat steps 6,7,8 and 9 until the pump runs properly
- PC11.** check the oil level at regular intervals
- PC12.** observe any slurry leakage during the pump operation and stop the pump immediately, if found

Stop the slurry pump unit

To be competent, the user/individual on the job must be able to:

- PC13.** close the outlet valve, but ensure that the pump runs in this condition for no more than a few seconds
- PC14.** stop the pump
- PC15.** switch off flushing and/or cooling/heating liquid supplies at a time appropriate to the process
- PC16.** maintain and complete all required documentation clearly, concisely as per SOP

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** quality and damage checks to be done and importance of the same
- KU3.** risk and impact of not following defined procedures/work instructions
- KU4.** functioning of slurry pump and its various components and accessories
- KU5.** normal running characteristics of slurry pump unit
- KU6.** SOP recommended by the manufacturer for operating slurry pump
- KU7.** how to start, operate and stop the slurry pump unit
- KU8.** how to maintain flow at inlet and outlet valves as per requirement
- KU9.** effect of improper flow, pressure, bearing temperature and oil levels on the equipment
- KU10.** general safety rules during slurry pump operation
- KU11.** operation and safety aspects of the equipment
- KU12.** possible causes of common problems during slurry pump operation and remedies
- KU13.** signages, mining area-specific signs and other safety and emergence signals

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** read and interpret symbols and readings
- GS3.** write work related information in English/regional language
- GS4.** communicate the work requirements to the supervisor and co-workers
- GS5.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS6.** make decisions pertaining to the concerned area of work
- GS7.** plan and organize the work order and tasks
- GS8.** recognise a workplace problem and take suitable action
- GS9.** follow instructions and work on areas of improvement identified
- GS10.** complete the assigned tasks timely
- GS11.** use reasoning skills to identify and resolve basic problems
- GS12.** analyze and detect any potential problems which could arise during operation
- GS13.** report to the supervisor or deal with a colleague individually, depending on the type of concern

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Start and operate the slurry pump unit</i>	24	37	-	14
PC1. ensure flushing and/or cooling liquid supplies are turned on and all vent connections are closed before starting the pump	2	3	-	1
PC2. open the inlet valve and fully close or partially open the outlet valve before starting the pump	2	3	-	1
PC3. open any recirculation or cooling lines	2	3	-	1
PC4. start the motor and slowly open the outlet valve until the pump reaches the desired flow	2	4	-	2
PC5. check the pressure gauge to ensure that the pump reaches the required outlet pressure	2	3	-	2
PC6. stop the pump, if there is no pressure or low pressure	2	3	-	1
PC7. perform the pump priming process again and restart the motor again	2	3	-	1
PC8. monitor the pump for bearing temperature, excessive vibration, and noise during the operation	2	3	-	1
PC9. shut down the pump immediately, if the pump exceeds normal levels and take the corrective action as per SOP	2	3	-	1
PC10. repeat steps 6,7,8 and 9 until the pump runs properly	2	3	-	1
PC11. check the oil level at regular intervals	2	3	-	1
PC12. observe any slurry leakage during the pump operation and stop the pump immediately, if found	2	3	-	1
<i>Stop the slurry pump unit</i>	6	13	-	6

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. close the outlet valve, but ensure that the pump runs in this condition for no more than a few seconds	2	3	-	2
PC14. stop the pump	2	4	-	2
PC15. switch off flushing and/or cooling/heating liquid supplies at a time appropriate to the process	1	3	-	1
PC16. maintain and complete all required documentation clearly, concisely as per SOP	1	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3222
NOS Name	Perform operation of slurry pump unit
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
NSQF Level	3.0
Credits	5
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	26/08/2027
NSQC Clearance Date	27/08/2024

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MIN/N3223: Perform maintenance of the slurry pump

Description

This OS unit is about performing routine maintenance and troubleshooting tasks on the slurry pump.

Scope

The scope covers the following :

- Prepare for maintenance work
- Repair and maintain the slurry pump
- Perform post-maintenance activities

Elements and Performance Criteria

Prepare for maintenance work

To be competent, the user/individual on the job must be able to:

- PC1.** identify the slurry pump manufacturer's specifications from the user manual and drawings
- PC2.** read maintenance schedule and checklist and plan the time and schedule for conducting the maintenance
- PC3.** identify and arrange the tools, consumables and spare parts required during the task
- PC4.** follow safety practices during maintenance activities as per organisational SOP

Repair and maintain the slurry pump

To be competent, the user/individual on the job must be able to:

- PC5.** perform basic health check-up of slurry pump unit as specified in the maintenance checklist
- PC6.** check the level and condition of the oil through the sight glass on the bearing frame
- PC7.** replenish coolants and lubricants as per SOP
- PC8.** clean dirt and dust from areas around close clearances, bearing housings and motors
- PC9.** check coupling alignment and re-align if necessary
- PC10.** check the slurry pump parts i.e. motor, impellers, casing, shaft and bearing assembly, shaft sleeves, seals, strainer, agitator, auxiliary supplies e.g. heating/cooling (if fitted) wiring harness, connectors etc. for proper functioning to find out root cause of the problems like any leakage, short circuit in parts, breakage of wires and clamps, etc. and discuss the same with senior if required
- PC11.** check for unusual noise vibration and bearing temperatures
- PC12.** check the pump and piping, seal chamber and stuffing box etc. for leaks
- PC13.** check any are functioning correctly
- PC14.** check the instrument panel and control gauge to identify out of range readings to find the defect/cause of failure
- PC15.** replace/change the spare parts and consumables of the slurry pump as per the schedule
- PC16.** ensure proper lock out and tag out of slurry pump to avoid any untoward incident triggered due to unknowingly operation of machine / system under maintenance

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PC17. assess when the problem is beyond his competence and report the problem to suitably qualified/designated and competent personnel

Perform post-maintenance activities

To be competent, the user/individual on the job must be able to:

PC18. conduct trial run of the slurry pump unit rig and verify that specified parameters are attained with no abnormalities

PC19. dispose off waste materials such as failed parts/aggregates, as per organisation's policies

PC20. fill the daily, weekly and monthly maintenance/defect sheets as per the format lay down by the organization

PC21. record all repairs carried out, time taken and unplanned tasks encountered during the maintenance activities

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. relevant standards and procedures followed in the company

KU2. quality and damage checks to be done and importance of the same

KU3. risk and impact of not following defined procedures/work instructions

KU4. functioning of slurry pump unit and its various systems and components

KU5. normal running characteristics of slurry pump unit

KU6. how to read maintenance schedules and checklists recommended by the manufacturer

KU7. how to read machine layout or drawing and wiring diagrams to understand its structure

KU8. Standard Operating Procedures (SOP) recommended by OEM for using tools and equipment required

KU9. how to collect and store consumables, spare parts, tools, etc. as per organizational procedures

KU10. use of appropriate PPE, material handling equipment and tools for completing the maintenance tasks

KU11. corrective actions for common faults and failures in slurry pump unit

KU12. documentation required regarding repair, maintenance and service performed

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read work instructions, equipment manuals and process documents

GS2. read and interpret symbols and readings

GS3. write work related information in English/regional language

GS4. communicate the work requirements to the supervisor and co-workers

GS5. effectively communicate, listen and comprehend the information given by various sources about the site

GS6. make decisions pertaining to the concerned area of work



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- GS7.** plan and organize the work order and tasks
- GS8.** recognise a workplace problem and take suitable action
- GS9.** follow instructions and work on areas of improvement identified
- GS10.** complete the assigned tasks timely
- GS11.** use reasoning skills to identify and resolve basic problems
- GS12.** analyze and detect any potential problems which could arise during operation
- GS13.** report to the supervisor or deal with a colleague individually, depending on the type of concern

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for maintenance work</i>	4	8	-	4
PC1. identify the slurry pump manufacturer's specifications from the user manual and drawings	1	2	-	1
PC2. read maintenance schedule and checklist and plan the time and schedule for conducting the maintenance	1	2	-	1
PC3. identify and arrange the tools, consumables and spare parts required during the task	1	2	-	1
PC4. follow safety practices during maintenance activities as per organisational SOP	1	2	-	1
<i>Repair and maintain the slurry pump</i>	20	32	-	13
PC5. perform basic health check-up of slurry pump unit as specified in the maintenance checklist	1	2	-	1
PC6. check the level and condition of the oil through the sight glass on the bearing frame	1	2	-	1
PC7. replenish coolants and lubricants as per SOP	1	2	-	1
PC8. clean dirt and dust from areas around close clearances, bearing housings and motors	1	2	-	1
PC9. check coupling alignment and re-align if necessary	2	2	-	1
PC10. check the slurry pump parts i.e. motor, impellers, casing, shaft and bearing assembly, shaft sleeves, seals, strainer, agitator, auxiliary supplies e.g. heating/cooling (if fitted) wiring harness, connectors etc. for proper functioning to find out root cause of the problems like any leakage, short circuit in parts, breakage of wires and clamps, etc. and discuss the same with senior if required	2	3	-	1
PC11. check for unusual noise vibration and bearing temperatures	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. check the pump and piping, seal chamber and stuffing box etc. for leaks	2	3	-	1
PC13. check any are functioning correctly	2	3	-	1
PC14. check the instrument panel and control gauge to identify out of range readings to find the defect/cause of failure	2	3	-	1
PC15. replace/change the spare parts and consumables of the slurry pump as per the schedule	2	3	-	1
PC16. ensure proper lock out and tag out of slurry pump to avoid any untoward incident triggered due to unknowingly operation of machine / system under maintenance	1	2	-	1
PC17. assess when the problem is beyond his competence and report the problem to suitably qualified/designated and competent personnel	1	2	-	1
<i>Perform post-maintenance activities</i>	6	10	-	3
PC18. conduct trial run of the slurry pump unit rig and verify that specified parameters are attained with no abnormalities	2	3	-	1
PC19. dispose off waste materials such as failed parts/aggregates, as per organisation's policies	2	3	-	1
PC20. fill the daily, weekly and monthly maintenance/defect sheets as per the format lay down by the organization	1	2	-	1
PC21. record all repairs carried out, time taken and unplanned tasks encountered during the maintenance activities	1	2	-	-
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3223
NOS Name	Perform maintenance of the slurry pump
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
NSQF Level	3.0
Credits	4
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	26/08/2027
NSQC Clearance Date	27/08/2024

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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2026
Next Review Date	18/02/2029
NSQC Clearance Date	18/02/2026

Qualification Pack

MIN/N1702: Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines

Description

This unit is about adhering to health, safety and environmental guidelines in Underground Metalliferous Mines (UMM) during the execution of various tasks and operations and maintenance.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work
- PC2.** comply with safety, health and security-related regulations/guidelines at the mine e.g. follow Standard Operating Procedure (SOP) for material handling in underground (U/G) mine
- PC3.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC4.** check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area
- PC5.** take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment
- PC6.** use self-rescue apparatus, appropriately when required
- PC7.** follow appropriate emergency response procedure during emergency such as fire, water inrush, fall of ground etc.
- PC8.** follow precautions against U/G electrical appliances
- PC9.** follow appropriate Standard Operating Procedure while working near any isolated and sealed off area of the mine
- PC10.** follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire
- PC11.** follow the manufacturer's instructions for care and safe operation of mine machinery and equipment
- PC12.** identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face
- PC13.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC14.** follow gas detecting alarm signal on leakage of inflammable gases and laid out procedure to ensure safety
- PC15.** follow process for reporting any unsafe act/condition in work area to the concerned person

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- PC16.** use underground mine communication system
- PC17.** ensure positive isolation near the work place if applicable
- PC18.** use appropriate PPE as per the requirement
- PC19.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC20.** disinfect the machine/tools before and after work/task
- PC21.** maintain hygiene at the work site
- PC22.** report any symptoms of illness to the shift-incharge
- PC23.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC24.** identify the environmental impact of mining related operations and follow steps to reduce those impact
- PC25.** follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery
- PC26.** ensure that the stowing practices produce minimum disturbance to the surface
- PC27.** ensure that the subgrade ore is carried out to surface and stacked separately at the earmarked place
- PC28.** ensure the productivity of the machine for material/fuel conservation
- PC29.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC30.** follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU2.** duties and rights of workers
- KU3.** selection process of person for rescue training
- KU4.** about Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR)
- KU5.** isolation and sealed off area of the mine
- KU6.** various types of gases found in the mine and their effect
- KU7.** self-rescue apparatus and their uses
- KU8.** provision of medical examination (IME & PME) of person employed as per Mines Rules 1955
- KU9.** importance of first aid and hygiene
- KU10.** importance of blind spot survey/training
- KU11.** about different types of machinery used in U/G mines
- KU12.** different types of supporting systems used in U/G mines as per SSP and SSR
- KU13.** about precautions to be taken when handling heavy equipment
- KU14.** various problems/incidents likely to occur
- KU15.** role of Internal Safety Organization, safety committee, workman's inspector and DGMS

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- KU16.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU17.** common sources of pollution in the mines and ways to minimize it
- KU18.** various types of fire extinguishers
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, Gas Detector, Safety Lamp, Self-Contained Breathing Apparatus, gum boots, ear plugs, Face Mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization as per DGMS guideline
- KU22.** mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** Take-5 (Personal Risk Assessment) training (DGMS Tech. circulars 2/2014)
- KU25.** rules and regulations for safety and security while handling hazardous materials
- KU26.** safety appliances and rescue equipment
- KU27.** importance and use of various communication system used in UG mines
- KU28.** importance of positive isolation at working site
- KU29.** safety and occupational health policy of organisation
- KU30.** six directional hazard identification process
- KU31.** basic personal and workplace hygiene
- KU32.** importance of FAB (Fresh Air Base)
- KU33.** basic provisions in Mines Creche Rules, 1966 (MCR) for females employed in the mines
- KU34.** the role and responsibilities of rescue room and rescue station and how to contact them in case of emergency
- KU35.** importance of taking shelter at the miner's station during blasting operation
- KU36.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU37.** importance of following infection control policies, '5-S' practices, and waste management
- KU38.** importance of water/material/energy conservation and management
- KU39.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read basic English language
- GS4.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS5.** use the digital information from machine
- GS6.** use basic applications of a computer
- GS7.** plan and organize the work order and tasks
- GS8.** use reasoning skills to identify and resolve fundamental problems

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GS9. complete the assigned tasks timely

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	24	42	-	17
PC1. undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work	-	2	-	2
PC2. comply with safety, health and security-related regulations/guidelines at the mine e.g. follow Standard Operating Procedure (SOP) for material handling in underground (U/G) mine	1	3	-	-
PC3. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	4	-	1
PC4. check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area	1	1	-	-
PC5. take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment	1	2	-	1
PC6. use self-rescue apparatus, appropriately when required	-	4	-	1
PC7. follow appropriate emergency response procedure during emergency such as fire, water inrush, fall of ground etc.	1	2	-	1
PC8. follow precautions against U/G electrical appliances	2	2	-	2
PC9. follow appropriate Standard Operating Procedure while working near any isolated and sealed off area of the mine	2	2	-	1
PC10. follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire	2	4	-	-
PC11. follow the manufacturer's instructions for care and safe operation of mine machinery and equipment	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face	1	1	-	-
PC13. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	-	-	-
PC14. follow gas detecting alarm signal on leakage of inflammable gases and laid out procedure to ensure safety	1	2	-	1
PC15. follow process for reporting any unsafe act/condition in work area to the concerned person	-	2	-	1
PC16. use underground mine communication system	-	1	-	-
PC17. ensure positive isolation near the work place if applicable	1	1	-	1
PC18. use appropriate PPE as per the requirement	2	4	-	2
PC19. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	1	-	1
PC20. disinfect the machine/tools before and after work/task	1	1	-	-
PC21. maintain hygiene at the work site	1	1	-	1
PC22. report any symptoms of illness to the shift-incharge	1	-	-	-
PC23. identify six directional hazards at workplace and take decisions accordingly	2	-	-	-
<i>Follow environmental guidelines</i>	6	8	-	3
PC24. identify the environmental impact of mining related operations and follow steps to reduce those impact	1	2	-	-
PC25. follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery	-	1	-	-
PC26. ensure that the stowing practices produce minimum disturbance to the surface	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure that the subgrade ore is carried out to surface and stacked separately at the earmarked place	1	-	-	1
PC28. ensure the productivity of the machine for material/fuel conservation	1	1	-	-
PC29. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	1	-	1
PC30. follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.	1	2	-	1
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1702
NOS Name	Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Underground, Specialized Operations, Electrical Services, Mechanical Services
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQF Clearance Date	25/02/2026

Qualification Pack

MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines

Description

This unit is about adhering to health, safety and environmental guidelines at the Open Cast Mines while executing various tasks, maintenance and operations.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow Environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** comply with safety, health, and security-related regulations/guidelines at the opencast mine
- PC2.** follow the safety instructions given by the workman's inspector
- PC3.** follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.
- PC4.** take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations
- PC5.** follow appropriate Safe Operating Procedure (SOP) while dealing with explosives
- PC6.** respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility
- PC7.** provide first aid to an injured person
- PC8.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC9.** use appropriate PPE as per the requirement
- PC10.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC11.** disinfect the machine/tools before and after work/task
- PC12.** maintain hygiene at the work site
- PC13.** report any symptoms of illness to the shift-incharge

Follow Environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC14.** identify the environmental impact of related opencast mining operations
- PC15.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC16.** ensure not to mix topsoil with waste in day to day tasks
- PC17.** ensure that HEMM is washed at the designated location

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- PC18.** ensure the productivity of the machine for material/fuel conservation
- PC19.** follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)
- PC20.** assist supervisor for reducing environmental impact caused due to related mining operations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** about various environmental awareness program related to mining, organized by the various government bodies/company
- KU2.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU3.** how to identify six-directional hazards
- KU4.** basic mining terminologies and definitions
- KU5.** importance of blind spot survey/training
- KU6.** means of access and egress from the mines, location of workshop, haul roads and working face including dump yards
- KU7.** duties of workers under The Mines act-1952
- KU8.** working hours and accident compensation as per The Mines act-1952
- KU9.** about precautions to be taken when handling heavy equipment
- KU10.** various problems/incidents likely to occur
- KU11.** hierarchy of the reporting
- KU12.** machine operation, condition of the machine and worksite
- KU13.** proper documents specific to the machine
- KU14.** role of workmen inspector, safety committee and internal safety organization
- KU15.** the process of top soil removal and management
- KU16.** mine sump and pumping system of the mines
- KU17.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU18.** common sources of pollution in the mines and ways to minimize it
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, gas detector, safety lamp, self-contained breathing apparatus, gum boots, ear plugs, face mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization
- KU22.** signages, mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** basic personal and workplace hygiene
- KU25.** importance of sensitization towards different genders and PWD (Persons with Disabilities)

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS4.** plan and organize the work order and jobs
- GS5.** use reasoning skills to identify and resolve fundamental problems
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	20	34	-	15
PC1. comply with safety, health, and security-related regulations/guidelines at the opencast mine	2	2	-	1
PC2. follow the safety instructions given by the workman's inspector	1	3	-	1
PC3. follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.	3	3	-	1
PC4. take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations	1	3	-	1
PC5. follow appropriate Safe Operating Procedure (SOP) while dealing with explosives	2	3	-	2
PC6. respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility	2	3	-	1
PC7. provide first aid to an injured person	2	3	-	1
PC8. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	3	-	1
PC9. use appropriate PPE as per the requirement	2	4	-	2
PC10. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	2	-	1
PC11. disinfect the machine/tools before and after work/task	1	2	-	1
PC12. maintain hygiene at the work site	1	1	-	1
PC13. report any symptoms of illness to the shift-incharge	1	2	-	1
<i>Follow Environmental guidelines</i>	10	16	-	5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. identify the environmental impact of related opencast mining operations	2	2	-	1
PC15. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	2	-	1
PC16. ensure not to mix topsoil with waste in day to day tasks	2	2	-	1
PC17. ensure that HEMM is washed at the designated location	2	2	-	1
PC18. ensure the productivity of the machine for material/fuel conservation	1	3	-	-
PC19. follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)	1	3	-	1
PC20. assist supervisor for reducing environmental impact caused due to related mining operations	1	2	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1703
NOS Name	Follow Health, Safety, and Environmental Guidelines for opencast mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, , Exploration, Mineral Estimation, Planning, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Opencast, Specialized Operations, Electrical Services, Mechanical Services, Instrumentation and Control Systems, Ore Processing
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQF Clearance Date	11/03/2026

Qualification Pack

MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines

Description

This unit is about adhering to health, safety and environmental guidelines in Underground Coal Mines (UCM) during execution of various tasks, operations and maintenance.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** follow preventive measures against firedamp, whitedamp, blackdamp etc.
- PC2.** use the flame safety lamp for detecting the methane gas as per Standard Operating Procedure (SOP)
- PC3.** undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work
- PC4.** comply with safety, health and security-related regulations/guidelines at the mine e.g. SOP for material handling in underground (U/G) mine
- PC5.** ensure that oil, grease, canvas or other inflammable material are stored in fire-proof receptacle
- PC6.** ensure that every instrument, apparatus and equipment are DGMS approved before these are used
- PC7.** ensure that Armoured face conveyor (AFC) and chocks must be kept in a straight line for every cycle of operations and tightened up to the setting pressure while keeping it in full contact with the roof, applicable for longwall mining
- PC8.** provide first aid to an injured person
- PC9.** follow safety precautions against spontaneous heating of the coal
- PC10.** operate various types of fire extinguishers to control different types of fire at worksite, if required
- PC11.** ensure that no person is traveling/working/staying under unsupported roof
- PC12.** check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area
- PC13.** take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment
- PC14.** use self-rescue apparatus appropriately when required
- PC15.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC16.** follow precautions against U/G electrical appliances

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- PC17.** take proper care against damage and accidents while loading, transporting, dismantling and erecting of roof supports
- PC18.** follow appropriate SOP while working near any isolated and sealed off area of the mine
- PC19.** ensure that the roof and the sidewalls of the mine face (or newly exposed area of the mines) have been scaled/dressed properly
- PC20.** take relevant safety precautions during depillaring operation in UCM
- PC21.** follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire
- PC22.** follow the manufacturer's instructions for care and safe operation of mine machinery and equipment
- PC23.** identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face
- PC24.** follow laid out SOP in case of alarm signal for leakage of inflammable gases
- PC25.** follow the process of reporting any unsafe act/condition in the working area to the concerned person
- PC26.** use underground mine communication system
- PC27.** ensure positive isolation near the work place if applicable
- PC28.** use appropriate Personal Protective Equipment (PPE) as per the requirement
- PC29.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC30.** disinfect the machine/tools before and after work/task
- PC31.** maintain hygiene at the work site
- PC32.** report any symptoms of illness to the shift-incharge
- PC33.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC34.** identify the environmental impact of mining related operations and follow steps to reduce those impact
- PC35.** follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery
- PC36.** ensure that the stowing practices produce minimum disturbance to the surface
- PC37.** ensure that the subgrade coal is carried out to surface and stacked separately at the earmarked place
- PC38.** ensure the productivity of the machine for material/fuel conservation
- PC39.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC40.** follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines specified by Directorate General of Mine Safety (DGMS)

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- KU2.** duties and rights of workers
- KU3.** selection process of person for rescue training
- KU4.** about Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR)
- KU5.** isolation and sealed off area of the mine
- KU6.** various types of gases available in the mine and their effects; and their control measures
- KU7.** self-rescue apparatus and their uses
- KU8.** provision of medical examination (Initial Medical Examination (IME) & Periodical Medical Examination (PME)) of a person employed, as per Mines Rules 1955
- KU9.** importance of first aid and hygiene
- KU10.** about different types of machineries used in U/G mines
- KU11.** different types of supporting system used in U/G mines as per SSP and SSR
- KU12.** about precautions to be taken when handling heavy equipment
- KU13.** various problems/incidents likely to occur
- KU14.** role of Internal Safety Organization, safety committee, workman's inspector and DGMS
- KU15.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU16.** common sources of pollution in the mines and ways to minimize it
- KU17.** various types of fire extinguishers
- KU18.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, Gas Detector, Safety Lamp, Self-Contained Breathing Apparatus, gum boots, ear plugs, Face Mask, etc.
- KU19.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU20.** mining area-specific signs, and other safety and emergency signals
- KU21.** the outcome of violation of safety procedures
- KU22.** Take-5 (Personal Risk Assessment) training (DGMS Tech. circulars 2/2014)
- KU23.** hazardous material safety, security rules and regulations
- KU24.** safety appliances and rescue equipment
- KU25.** importance and use of various communication system used in UG mines
- KU26.** importance of positive isolation at working site
- KU27.** safety and occupational health policy of organisation
- KU28.** six directional hazard identification process
- KU29.** importance of blind spot survey/training
- KU30.** basic personal and workplace hygiene
- KU31.** importance of FAB (Fresh Air Base)
- KU32.** basic provisions in Mines Creche Rules, 1966 (MCR) for any females employed in the mines
- KU33.** about basic safety regulations of Coal Mines Regulation, 2017 (CMR)
- KU34.** types of stone dust barrier and its importance
- KU35.** coal dust explosion and its preventive measures
- KU36.** classification of coal mines as per the degree of gassiness of coal seams such as first degree, second degree, and third-degree mines
- KU37.** precautions as per the gassiness of the coal mines

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- KU38.** use of flame safety lamp and its parts
- KU39.** about coal mines occupational disease such as pneumoconiosis or 'black lung' and their preventive measures
- KU40.** Standard of Ventilation as per the section 153 of the CMR 2017
- KU41.** Standard of Lighting as per the section 178 of the CMR 2017
- KU42.** the roles, duties and responsibilities of rescue team members, rescue room and rescue station and how to contact them in case of emergency
- KU43.** the correct steps for conducting any rescue work as per Mine Rescue Rule (MRR)
- KU44.** importance of taking shelter at the miner's station during blasting operation
- KU45.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU46.** importance of following infection control policies, '5-S' practices, and waste management
- KU47.** importance of water/material/energy conservation and management
- KU48.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read basic English language
- GS4.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS5.** use the digital information from a machine
- GS6.** plan and organize the work order and tasks
- GS7.** use basic applications of the computer
- GS8.** use reasoning skills to identify and resolve fundamental problems
- GS9.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	24	43	-	14
PC1. follow preventive measures against firedamp, whitedamp, blackdamp etc.	1	1	-	-
PC2. use the flame safety lamp for detecting the methane gas as per Standard Operating Procedure (SOP)	-	1	-	-
PC3. undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work	1	1	-	1
PC4. comply with safety, health and security-related regulations/guidelines at the mine e.g. SOP for material handling in underground (U/G) mine	1	1	-	1
PC5. ensure that oil, grease, canvas or other inflammable material are stored in fire-proof receptacle	-	1	-	-
PC6. ensure that every instrument, apparatus and equipment are DGMS approved before these are used	1	1	-	-
PC7. ensure that Armoured face conveyor (AFC) and chocks must be kept in a straight line for every cycle of operations and tightened up to the setting pressure while keeping it in full contact with the roof, applicable for longwall mining	3	3	-	-
PC8. provide first aid to an injured person	1	2	-	1
PC9. follow safety precautions against spontaneous heating of the coal	1	1	-	-
PC10. operate various types of fire extinguishers to control different types of fire at worksite, if required	-	2	-	-
PC11. ensure that no person is traveling/working/staying under unsupported roof	1	-	-	-
PC12. check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area	-	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment	-	1	-	1
PC14. use self-rescue apparatus appropriately when required	-	2	-	-
PC15. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	1
PC16. follow precautions against U/G electrical appliances	2	2	-	1
PC17. take proper care against damage and accidents while loading, transporting, dismantling and erecting of roof supports	1	2	-	-
PC18. follow appropriate SOP while working near any isolated and sealed off area of the mine	1	1	-	1
PC19. ensure that the roof and the sidewalls of the mine face (or newly exposed area of the mines) have been scaled/dressed properly	1	1	-	-
PC20. take relevant safety precautions during depillaring operation in UCM	-	2	-	1
PC21. follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire	1	1	-	-
PC22. follow the manufacturer's instructions for care and safe operation of mine machinery and equipment	-	1	-	1
PC23. identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face	-	1	-	1
PC24. follow laid out SOP in case of alarm signal for leakage of inflammable gases	1	1	-	-
PC25. follow the process of reporting any unsafe act/condition in the working area to the concerned person	-	1	-	1
PC26. use underground mine communication system	-	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure positive isolation near the work place if applicable	1	1	-	-
PC28. use appropriate Personal Protective Equipment (PPE) as per the requirement	2	2	-	1
PC29. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	-	1	-	-
PC30. disinfect the machine/tools before and after work/task	1	1	-	-
PC31. maintain hygiene at the work site	1	1	-	-
PC32. report any symptoms of illness to the shift-incharge	1	1	-	-
PC33. identify six directional hazards at workplace and take decisions accordingly	-	1	-	1
<i>Follow environmental guidelines</i>	6	7	-	6
PC34. identify the environmental impact of mining related operations and follow steps to reduce those impact	1	1	-	1
PC35. follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery	1	1	-	1
PC36. ensure that the stowing practices produce minimum disturbance to the surface	1	1	-	-
PC37. ensure that the subgrade coal is carried out to surface and stacked separately at the earmarked place	1	1	-	1
PC38. ensure the productivity of the machine for material/fuel conservation	1	1	-	1
PC39. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	-	1	-	1
PC40. follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.	1	1	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1704
NOS Name	Follow Health, Safety, and Environmental guidelines for underground coal mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Underground, Specialized Operations, Electrical Services, Mechanical Services
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

MIN/N1705: Follow Health, Safety and Environmental guidelines for Rare Earth (RE) Chemical plant

Description

This unit is about adhering to health, safety and environmental guidelines critical in RE chemical plants

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** comply with occupational health and safety regulations adopted by the employer
- PC2.** follow the correct safety steps in case of accident or major failure
- PC3.** operate various grades of fire extinguishers
- PC4.** follow general safety precautions while handling cables; working near electrical installations, overhead lines and while working with various electrical equipment in the plant
- PC5.** provide first-aid to an injured person
- PC6.** use appropriate PPE as per the requirement
- PC7.** identify the hazards and risks
- PC8.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC9.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC10.** disinfect the site/panel/tools before and after work/task
- PC11.** maintain hygiene at the work site
- PC12.** report any symptoms of illness to the shift-incharge
- PC13.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC14.** identify the environmental impact of operations related to RE chemical plant and take steps to reduce the impact
- PC15.** ensure the productivity of the machine for material/fuel conservation
- PC16.** follow the process for collecting, storing, and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines and safety guidelines as prescribed by regulatory authorities like Atomic Energy Regulatory Board (AERB), United Nations Development Group (UNDG), etc.

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** the outcome of violation of safety procedures
- KU2.** duties and rights of workers
- KU3.** process for reporting any unsafe act/condition in work area
- KU4.** different types of RE chemical factories and detail of the plant one is working in
- KU5.** fencing, guarding, spillage control, etc.
- KU6.** importance of first aid and hygiene
- KU7.** importance of blind spot survey/training
- KU8.** provision of wages, working hours and accident compensation as per Atomic Energy Factory Rules
- KU9.** various types of chemical processes carried out in the plant
- KU10.** RE chemical plant safety procedures
- KU11.** about installation and handling of safety devices
- KU12.** code of practice for safe handling and transport of dangerous material and heavy equipment
- KU13.** safety and occupational health policy of organisation
- KU14.** basic personal and workplace hygiene
- KU15.** six directional hazard identification process
- KU16.** Internal Safety Organisation and role of safety committee, workman's inspector and AERB etc.
- KU17.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU18.** importance of following infection control policies, '5-S' practices, and waste management as prescribed by regulatory authorities like Atomic Energy Regulatory Board (AERB), United Nations Development Group (UNDG), etc.
- KU19.** importance of water/material/energy conservation and management
- KU20.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's job role
- GS2.** make decisions on the concerned area of work
- GS3.** read and explain manuals, health, and safety instructions, memos, etc.
- GS4.** plan and organize the work order and jobs
- GS5.** use reasoning skills to identify and resolve fundamental problems
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	26	44	-	17
PC1. comply with occupational health and safety regulations adopted by the employer	2	3	-	1
PC2. follow the correct safety steps in case of accident or major failure	2	4	-	1
PC3. operate various grades of fire extinguishers	2	4	-	1
PC4. follow general safety precautions while handling cables; working near electrical installations, overhead lines and while working with various electrical equipment in the plant	3	4	-	2
PC5. provide first-aid to an injured person	3	4	-	2
PC6. use appropriate PPE as per the requirement	2	5	-	2
PC7. identify the hazards and risks	3	6	-	1
PC8. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	3	5	-	2
PC9. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	1	-	1
PC10. disinfect the site/panel/tools before and after work/task	2	2	-	1
PC11. maintain hygiene at the work site	1	2	-	1
PC12. report any symptoms of illness to the shift-incharge	1	2	-	1
PC13. identify six directional hazards at workplace and take decisions accordingly	1	2	-	1
<i>Follow environmental guidelines</i>	4	6	-	3
PC14. identify the environmental impact of operations related to RE chemical plant and take steps to reduce the impact	1	2	-	1
PC15. ensure the productivity of the machine for material/fuel conservation	2	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. follow the process for collecting, storing, and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines and safety guidelines as prescribed by regulatory authorities like Atomic Energy Regulatory Board (AERB), United Nations Development Group (UNDG), etc.	1	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1705
NOS Name	Follow Health, Safety and Environmental guidelines for Rare Earth (RE) Chemical plant
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	27/08/2024
Next Review Date	26/08/2027
NSQC Clearance Date	27/08/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N3221.Prepare slurry pump unit for operation	30	50	0	20	100	15
MIN/N3222.Perform operation of slurry pump unit	30	50	0	20	100	25
MIN/N3223.Perform maintenance of the slurry pump	30	50	0	20	100	25
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	110	180	-	60	350	75

Elective: 1 Underground Metal

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1702.Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines	30	50	0	20	100	25
Total	30	50	-	20	100	25

Elective: 2 Opencast

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1703.Follow Health, Safety, and Environmental Guidelines for opencast mines	30	50	0	20	100	25
Total	30	50	-	20	100	25

Elective: 3 Underground Coal

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1704.Follow Health, Safety, and Environmental guidelines for underground coal mines	30	50	0	20	100	25
Total	30	50	-	20	100	25

Elective: 4 Rare Earth

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1705.Follow Health, Safety and Environmental guidelines for Rare Earth (RE) Chemical plant	30	50	0	20	100	25
Total	30	50	-	20	100	25

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.